

Aakriti Shah

[🌐 website](#)
[✉ u2aakriti@gmail.com](mailto:u2aakriti@gmail.com)
[📞 +1\(650\)867-0540](tel:+1(650)867-0540)
[🌐 aakritishah](#)
[in aakriti-shah-](#)

Research Interests

I am passionate about leveraging machine learning for social good and building systems that contribute to a **better society**. My research interests span **Natural Language Processing**, **LLM interpretability**, machine unlearning, **trustworthy Machine Learning**, and medical imaging.

Education

MS	University of Southern California , Computer Science	Aug 2024 - Expected May 2026
	Coursework: Trustworthy Machine Learning, Natural Language Processing, Algorithms, Text as Data (NLP and LLMs)	
BA	Rollins College , Computer Science & Business Management (<i>Honors</i>)	Sept 2020 – Dec 2023
	Coursework: Operating Systems, Algorithms, Software Engineering, Computer Organization & Architecture, Data Science & Analytics, Discrete Math, Software Engineering	

Honors & Awards

Dean's List , University of Southern California (multiple semesters)	2024-2025
Magna Cum Laude , Rollins College	2023
NSF REU Awardee , Pennsylvania State University	2023
Archibald Granville Bush Award: Excellent Standard in Academic Pursuit , Rollins College	2023
Honors Degree Program , Rollins College	2020-2023
President's List , Rollins College (multiple semesters)	2020-2023

Publications & Submissions

Shah, A. , & Le, T. (2025). " <i>The Limits of Obliviate: Evaluating Unlearning in LLMs via Stimulus-Knowledge Entanglement-Behavior Framework</i> " arXiv preprint arXiv:2510.25732 Submitted to EACL 2026/ACL RR 2026 in October 2025. <i>Decision expected January 2026.</i>	November 2025 Under Review
Shah, A. (presenting author), Bogale, Y., Murugan, N., Nair, V., Collins, K., Bahler, C., Feldman, M., Shiradkar, R. " <i>Deep-learning-based segmentation of glands and stroma on digitized H&E-stained prostate whole slide images</i> " Accepted abstract at SPIE Medical Imaging 2026 , February 2026. <i>Manuscript due January 2026.</i>	August 2025 Accepted for Oral Presentation
Shah, A. (presenting author), Bogale, Y., Nair, V., Collins, K., Bahler, C., Feldman, M., Shiradkar, R. " <i>Automated Gleason Pattern Segmentation in Prostate Whole Slide Digitized Pathology Using Foundation Model Feature Embeddings</i> " Submitted to AUA 2026 in November 2025.	November 2025 Under Review
Shah, A. (presenting author), Bogale, Y., Murugan, N., Nair, V., Collins, K., Bahler, C., Feldman, M., Shiradkar, R. " <i>Automated Segmentation of Gland and Stromal Morphology on Digitized H&E-Stained Prostate Whole Slide Images</i> " Submitted to AUA 2026 in November 2025.	November 2025 Under Review
Shah, A. (2023). " <i>Evaluating AI Sentiment Analysis</i> ." Rollins College Honors Program Theses, No. 218. Available at: scholarship.rollins.edu/honors/218/	Dec 2023 Undergraduate Honors Thesis

Research Experience

Research Assistant , ARKAI Lab - Dr. Thai Le (Computer Science, Indiana University)	May 2025 – Ongoing Project Paper
- Investigating targeted unlearning in LLMs through cognitive science frameworks, exploring parallels between machine unlearning and human memory retrieval. - Applying theories from psychology to analyze knowledge entanglement in unlearned	

LLMs, examining how unlearning affects associated concepts and resistance to **adversarial extraction attacks**.

- Developing evaluation frameworks to assess unlearning **robustness** and inform the design of more **secure LLMs**.

Research Assistant, AIMed Lab - Dr. Rakesh Shiradkar (Computer Science, Indiana University)

May 2024 – Ongoing
[Project](#) [Abstract](#)

- Conducting computational pathology research on prostate **cancer detection** through automated tissue compartment and Gleason pattern segmentation in H&E-stained whole slide images.
- Developed **novel postprocessing pipeline** incorporating lumen reconstruction to improve segmentation accuracy of tissue compartments (stroma, lumen, epithelium), achieving average Dice score of 0.895.
- Implementing and evaluating **deep learning and vision foundation architectures** (UNet, nnUNet, UNI) for robust multi-class segmentation across diverse histopathological datasets.
- Extending research to Gleason pattern grading (patterns 3, 4, 5) to enable deep learning automated cancer aggressiveness assessment.

Research Assistant, Dr. Anita Penkova (Computer Science, University of Southern California)

Sept 2024 – Ongoing

- Conducting **computer vision** research for early and automated detection of Diabetic Retinopathy in retinal fundus images.
- Developing and optimizing preprocessing pipelines (CLAHE, Gaussian Blur, MGA-CSG) to address limited high-quality training data and **improve model generalization** across diverse image quality conditions.

Research Project, Natural Language Processing

Jan 2025 – May 2025
[Project](#) [Paper](#)

- Developed Hinglish **conversational chatbot** by curating multilingual datasets with code-switched and synthetically generated training samples to address low-resource language challenges.
- **Fine-tuned instruction-tuned LLMs** for open-domain dialogue generation; evaluated outputs using syntactic and semantic metrics to assess code-switching quality and conversational coherence.

Honors Thesis, Dr. Daniel Myers

Jan 2023 – Dec 2023
[Project](#) [Paper](#)

- **Evaluated LLM performance** on sentiment analysis using manually labeled real-world interview data from non-profit Crave, comparing human-coded emotional assessments to model-generated outputs.
- Investigated **limitations of LLMs in capturing nuanced emotional context** in low-resource, qualitative settings, providing insights for responsible deployment in social science research.

Research Intern, National Science Foundation REU

Jan 2023 – Dec 2023
[Paper](#)

- Investigated **adversarial robustness of ML models** for autonomous driving systems, analyzing vulnerabilities under perturbation attacks and evaluating transfer learning across architectures.
- Conducted interdisciplinary research, covering qualitative and quantitative methods, research ethics, and perspectives from computer science and psychology.

Technical Projects

voteSmarter, Civic Engagement Mobile App

Feb 2024 – June 2024
[Project](#)

- Developed mobile application connecting users to political candidates based on values alignment and location; integrated features for voter registration, polling locations, and candidate databases.

ASLearning, ML for Accessibility

Jan 2023 – May 2023

- Built mobile application using real-time object detection (TensorFlow) to facilitate communication between hearing-impaired and English-speaking users through sign language recognition.

[Project](#)

Technical Skills

Programming: Python, SQL, Java, JavaScript, R, Assembly (ARM), C, C++, XML, Dart, C#, LaTeX, Swift

ML/DL Frameworks: PyTorch, TensorFlow, Hugging Face, scikit-learn, pandas, NumPy

Tools & Platforms: Git, Docker, Jupyter, Google Cloud Platform, MongoDB, Flask, React, Node.js

Research Skills: Experiment design, data annotation, literature review, statistical analysis, scientific writing

Teaching and Outreach

Volunteer, Viterbi Impact

Aug 2024 – Ongoing

- Engaging in community service projects including fire clean-up, premature baby care, and food distribution.
- Helped distribute food for people who were affected by the 2025 Southern California Wildfires and the nationwide EDT crisis.

Volunteer, AAPI Coming Together (ACT)

May 2024 – Ongoing

- Supporting civic outreach and cultural advocacy within the Asian American community.
- Developed **voteSmarter** to support voter education efforts and civic engagement.

Instructor, iDTech Camps

June 2022 – Aug 2022

- Taught Java, Python, C#, and game development to K-12 students, promoting computational thinking and creative problem-solving in early learners.